

State of the Art in Technology: AI Top Resources

Math

[3Blue1Brown](#) (see for [linear algebra](#), [calculus](#))

[Khan Academy](#) (see for [calculus](#), [multivariable calculus](#))

[MIT OpenCourseWare](#) (see for [linear algebra](#), [multivariable calculus](#))

[Self Study Guide](#) (see for [linear algebra](#), [multivariable calculus](#), [abstract algebra](#))

[Seeing Theory](#) (see for [probability](#), [statistics](#))

Machine Learning

[3Blue1Brown](#) (see for [neural networks](#))

[Distill](#) (see articles on [handwriting](#), [features](#), [gaussians](#), [bayes](#), [polysemanticity](#))

[Transformers](#) (see articles on [architecture](#), [induction heads](#), [superposition](#))

[Two Minute Papers](#) (see for news)

[AI Explained](#) (see for commentary)

[fast.ai](#) (full course on deep learning)

Roadmap to Learn Math for Machine Learning: Calculus and Linear Algebra

1. *Lockdown Math* by 3Blue1Brown, Episodes 1, 2, 3, 4, 5, 6, 7

- Practice: Khan Academy Precalculus Unit on Complex Numbers, Lessons 1, 2, 5, 6
- Practice Khan Academy Algebra II Unit on Rational Exponents, Lessons 1, 2 (fine to skip challenge exercise); Unit on Exponential Models, Lessons 1, 2; Unit on Logarithms, Lessons 1, 2, 3 (up to properties of logarithms); Polynomials, Lessons 3, 4, 5.

2. *Essence of Calculus* by 3Blue1Brown, Episodes 1, 2, 3, 4, 5, 8

- Practice: Khan Academy AP Calculus BC Unit on Differentiation, Lessons 5, 6, 7, 8, 9; Unit on Differentiation (composite), Lesson 1 (chain rule intro).

3. *Multivariable Calculus on Khan Academy*

- Practice: Khan Academy Multivariable Calculus Unit 1, Lesson "Vectors and Matrices" by Laker Newhouse; Unit 2, Lessons 1, 2; Unit 3, Lesson 4 (article on gradient descent).

4. *Essence of Linear Algebra* by 3Blue1Brown

- Practice: Khan Academy Precalculus Unit on Vectors, Lessons 1, 2, 3, 4, 5
- Practice: fun with eigenvectors and change of basis (see problem at end of Chapter 14).
- Additional resource: *Linear Algebra Done Right* by Sheldon Axler (text and exercises; skip sections 3E and 3F on first pass).